Shaik Zaheer

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PROFESSIONAL SUMMARY

AI/ML Engineer with **1+ years of experience** n developing and deploying data-driven solutions that significantly improve business outcomes. Proficient in leveraging **advanced AI technologies**, including **Generative AI** and implementing **Retrieval-Augmented Generation (RAG)** models to engineer scalable, high-performance applications. Skilled in the **end-to-end implementation of machine learning algorithms**, particularly **classification and regression**, to drive process automation and enhance critical decision-making. Committed to creating robust, efficient systems while continuously integrating **emerging technologies** to deliver innovative, business-aligned solutions.

SKILLS

Programming Languages: Python (Pandas, NumPy, Scikit-learn), SQL, Java
Generative AI & LLMs: RAG, LangChain, Agents, LlamaIndex, Hugging Face, Ragas, GPT-4o, GPT-4, GPT-3.5 Turbo, Gemini 1.5 Pro, Mistral
Prompt Engineering: Zero-Shot, One-Shot, Few-Shot Learning, Chain of Thought, ReAct Prompting
Machine Learning & Deep Learning: Classification, Regression, Clustering, Text Classification, Named Entity Recognition, Transformers, RNNs, CNNs, ANNs.
Cloud Platforms & Services:
Azure: Azure OpenAI, Azure Pipelines
DevOps & MLOps Tools: Docker, Kubernetes, Helm, Jenkins, GitLab CI, Terraform, Ansible, Prometheus, Grafana, ELK Stack, SonarQube, Aqua Security
Vector Databases: Pinecone, Chroma DB,Weavite,Quadrant.
Frameworks & Tools: FastAPI, Flask, TensorFlow, React, DBT
Infrastructure & Automation: CI/CD Pipeline Design, Microservices Deployment, Cloud Automation, Infrastructure as Code (IaC), Monitoring & Logging, Cloud Security & Compliance

PROFESSIONAL EXPERIENCE

AI/ML Engineer, Value Momentum **12/2024 – Present | Hyderabad, India**

- Built an enterprise-grade knowledge base system using Doxygen and Graphviz**,** assisted by GitHub Copilot, to auto-generate technical documentation, architecture diagrams, and dependency graphs for internal engineering teams.
- Created automated documentation pipelines to convert project specifications into structured, searchable technical artifacts, improving documentation accuracy and accessibility by 50%.
- Designed and developed a RAG-based enterprise chatbot that leverages the knowledge base to provide accurate project-specific insights, significantly reducing onboarding time and improving internal knowledge discovery.
- Developed the front-end chatbot UI using JavaScript, Tailwind CSS, and modular components, ensuring a clean, modern, and responsive user experience.
- Deployed the complete chatbot system to Azure using Azure App Services, Azure Storage, and CI/CD pipelines for secure, scalable, and cloud-native performance.
- Implemented LLM-powered semantic search and context retrieval, enabling engineers to quickly locate information across repositories, diagrams, and architectural documentation.
- Collaborated with architects and subject-matter experts to design metadata tagging, Graphviz-based relationship mapping, and continuous documentation updates.
- Conducted internal training on LLM development, documentation automation using Doxygen, RAG pipeline design, and AI-assisted engineering workflows, accelerating GenAI adoption within teams.

PROJECTS

Enterprise Knowledge Base & RAG Chatbot System

Technologies: Python, Doxygen, Graphviz, GitHub Copilot, FastAPI, LangChain, RAG, Tailwind CSS, JavaScript, Azure App Services, Azure Storage, Vector DB (FAISS/Chroma), Prompt Engineering

- Built an automated **enterprise knowledge base system** using **Doxygen** and **Graphviz** to generate complete technical documentation, API references, architecture diagrams, class hierarchies, and dependency graphs from large codebases.
- Leveraged **GitHub Copilot** to accelerate documentation creation, ensuring highly structured .dox files and diagram generation for engineers.
- Implemented a **RAG-based chatbot** that retrieves context from the Doxygen-generated knowledge base, enabling engineers to instantly access project details, architecture flows, and configuration references.
- Developed a modern, responsive **chatbot front-end** using **JavaScript + Tailwind CSS**, providing a clean UI with smooth interactions and developer-friendly accessibility.
- Integrated the chatbot with **FastAPI backend**, enabling secure API communication, document ingestion, embedding pipelines, and vector search.
- Deployed the full application on **Azure** using Azure App Services and Azure Storage, ensuring scalability, reliability, and CI/CD integration for continuous updates.
- Implemented **semantic search, metadata tagging, and relationship mapping**, allowing engineers to find information across UML diagrams, code modules, and documentation instantly.
- Delivered internal demos and training sessions on **Doxygen automation, Graphviz visualization, and RAG best practices**, supporting organization-wide adoption of AI-assisted documentation tools.

Azure GenAI Multi-Agent Cloud Governance Platform

Technologies: Azure OpenAI, LangChain, FastAPI, Terraform (Azure Provider), Azure Functions, Azure Cost Management API, Cosmos DB, Azure Monitor, Azure RBAC, Azure Policy, Python

- Architected a multi-agent cloud automation platform using **LangChain agents** orchestrated with Azure OpenAI for natural language–driven infrastructure provisioning.
- Designed four specialized agents (Requirement Understanding, Cost Estimation, Provisioning, Decommissioning) coordinated via LangChain agent executor.
- Implemented structured output parsing in LangChain to convert user prompts into validated Azure VM configurations (region, VM size, GPU, duration).
- Integrated Azure Cost Management APIs within a LangChain tool to dynamically estimate projected cloud spend before deployment.
- Automated Infrastructure-as-Code provisioning using Terraform with enforced tagging, RBAC, and Azure Policy compliance.
- Built timer-triggered Azure Functions for automated decommissioning to prevent idle resource cost leakage.
- Stored agent decisions, provisioning metadata, and audit logs in Cosmos DB for governance and traceability.
- Implemented risk-based approval workflow for high-cost or GPU provisioning requests using agent decision scoring.

EDUCATION

Teegala Krishna reddy Engineering College , Hyderabad-(May 2024)

- Bachelor of Technology in Computer Science and Engineering

Certifications

- Github Copilot Certification – 2025 – 2028
- LLM Fine tuning certification.